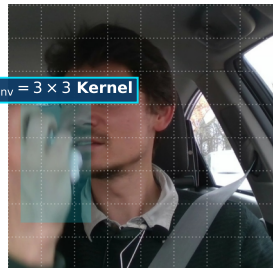
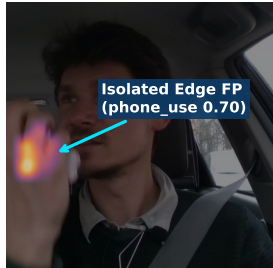


PARADIGM 1: PURE CONVOLUTION (YOLO26n / RepConv) — Strictly Local Spatial Receptive Fields

(a1) Spatial Grid & R_{conv}



(a2) Localized Activations



(a3) Backprop $\nabla_W \mathcal{L}_{\text{cls}}^{\text{neg}}$



Loss Dynamics (Pure RepConv) :

- Negative loss $\mathcal{L}_{\text{det}}^{\text{neg}}$ updates strictly localized sliding-window kernels (3×3).
- High-frequency edge noise (knuckles, rim) recurs uniformly across natural driving.
- **Early Saturation** : $r = 40\%$ random negatives saturate local filter silence ($99.1 \rightarrow 5.5$).
- **Hard Mining** : 0.0% benefit ($5.50 \rightarrow 5.50$).

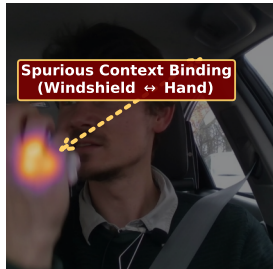
RULE: SAMPLE FOR CONVOLUTIONS
Uniform Random Sampling is Sufficient

PARADIGM 2: LINEAR AREA ATTENTION (YOLO12n / A2C2f) — Global Area-Spanning Attention Affinities

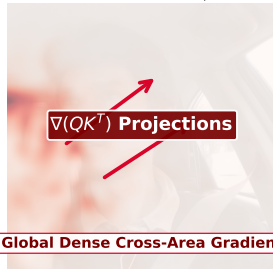
(b1) Area Partition (QK^T)



(b2) Area-Spanning Heatmap



(b3) Backprop $\nabla_{W_Q, W_K} \mathcal{L}_{\text{cls}}^{\text{neg}}$



Loss Dynamics (Area Attn) :

- Negative loss $\mathcal{L}_{\text{det}}^{\text{neg}}$ backpropagates globally across bipartite QK^T projections.
- Spurious bindings span distant cabin zones (glare ↔ hand ↔ headrest).
- **High Entropy Needed** : Random frames have plain background; lack cross-area signals.
- **Hard Mining** : **-65.2%** FP drop ($18.1 \rightarrow 6.3$).

RULE: CURATE FOR ATTENTION
Hard-Negative Curation is Essential